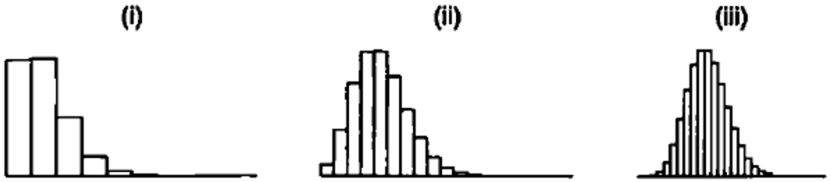
Chapter 9: One- and Two-Sample Estimation

Review

Shown below are probability histograms for the sum of 100, 400, and 900 draws from the box $\boxed{99 \quad \boxed{0} \text{'s} \quad \boxed{1}}$. Which histogram is which?



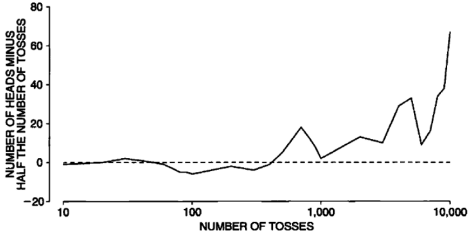
Review

You are drawing at random from a large box of red and blue marbles.

1. What distribution would be used to describe the expected number of red balls that you draw if you drew with replacement? What about without replacement?
2. As the number of draws (n) increases, the SD of the total number of reds in the sample goes _____
   (a) Up (b) Down
3. As the number of draws (n) increases, the SD of the average number of reds per draw goes _____. Hint: use your previous answer.
   (a) Up (b) Down

Review

PERCENTAGE OF HEADS − 50%
10
5
0
−5
−10
10
100
1,000
10,000
NUMBER OF TOSSES


NUMBER OF HEADS MINUS
HALF THE NUMBER OF TOSSES
80
60
40
20
0
−20
10
100
1,000
10,000
NUMBER OF TOSSES

Example

$$(110 \pm 3) - (90 \pm 4) = 20 \pm \underline{\hspace{3cm}}$$

Hint 1: $\pm$ refers to standard deviation. Hint 2: Think about how variances combine for independent quantities.

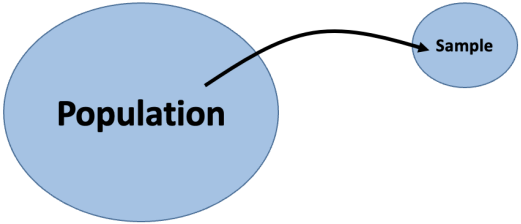
Learning Objectives

- Statistical inference
- Classical methods of estimation
- Single sample: estimating the mean
- Standard error of a point estimate
- Prediction interval
- Two samples: difference between two means
- Paired observations
- Single sample: estimating the variance
- Two samples: estimating the ratio of two variances

Introduction

- Build on concepts from previous chapters:
  - Sample properties of $\bar{X}$ and S^2 (Ch. 8)
  - Central Limit Theorem (Ch. 8)
  - Linear combinations of normal r.v.'s (Thm. 7.11)
  - Displaying data (Ch. 1)
- In this chapter:
  - Purpose of statistical inference
  - Estimation of population parameters

Statistical Inference



- Methods to make inferences (generalizations) about a population
- Inferences are based on information from a random sample
- Steps:
  1. Take a random sample from the population of interest
  2. Apply sampling distribution theory
  3. Make inference (estimates) about the population

Classical Methods of Estimation

- Take a sample, compute sample mean and sample variance
- These are *point estimates*
- Point estimate: a single number regarded as a sensible value for the unknown parameter
- Point estimation concepts:
  - Notation
  - Unbiasedness (expectation)
  - Efficiency (variance)

Point Estimators: Notation

- θ: parameter of interest
- $\hat{\Theta}$: estimator (a statistic / r.v.)
- $\hat{\theta}$: the realized value (computed from a sample)
- Example: population is $N(\mu, 1)$ where μ unknown
  - $\theta = \mu$
  - $\bar{X}$ can be an estimator
  - $\bar{x}$ is the observed sample mean
- Since $\hat{\Theta}$ is a r.v., we study its properties.

Point Estimators: Unbiased Estimators

- Which estimators are closest to the true mean?
- We can't know without knowing the true mean, so instead ask:
- Which estimator, across repeated samples, tends to be closest to the true mean?
- Consider an example...

Unbiased Estimators: Example (Epoxy Resin)

- 20 observations on dielectric breakdown voltage (epoxy resin):

  24.46, 25.61, 26.25, 26.42, 26.66, 27.15, 27.31, 27.54, 27.74, 27.94,
  27.98, 28.04, 28.28, 28.49, 28.50, 28.87, 29.11, 29.13, 29.50, 30.88

- Normal probability plot is straight $\Rightarrow$ assume normal.
- What is a "good" estimate of the mean? Why?

Unbiased Estimators

- We look for estimators that are unbiased.
- A statistic $\hat{\Theta}$ is an unbiased estimator of θ if

$$\mathbb{E}(\hat{\Theta}) = \theta \quad \text{for every possible value of } \theta.$$

- Bias of $\hat{\Theta}$ is $\mathbb{E}(\hat{\Theta}) - \theta$.
- Check: $\mathbb{E}(\bar{X}) = \mu$.

Variance of a Point Estimator

- What if you have several unbiased estimators?
- Choose the one with the smallest variance.
- This is called the *most efficient* estimator of θ.
- In practice, we often compare estimators using mean squared error (MSE).

Summary (Point Estimation)

- When choosing among estimators of θ, prefer one that is:
  - Unbiased, and
  - Most efficient (smallest variance)
- In this course, the common estimators we use can be shown to be efficient.
- Unbiased estimators to remember:
  - Mean μ:
  - Variance σ^2:

Interval Estimation

- $\bar{X}$ is a point estimate of μ.
- But a single value doesn't show how accurate it is.
- Better: report an interval of plausible values for the parameter.

Interval Estimation (Form)

- An interval estimate of θ has the form

$$\hat{\theta}_L < \theta < \hat{\theta}_U.$$

- Example: true average GPA of all freshman engineering students at Pitt.
- A sample might yield an interval 3.2 to 3.9.
- Larger n typically $\Rightarrow$ more accurate variance estimate $\Rightarrow$ shorter interval.

Confidence Intervals

- Different samples give different intervals.
- We can specify the probability the interval contains the true parameter:

$$P(\hat{\Theta}_L < \theta < \hat{\Theta}_U) = 1 - \alpha.$$

- The computed interval is a $100(1-\alpha)\%$ *confidence interval*.
- Endpoints are *confidence limits*.

So Far

- Two types of estimates:
  - Point estimates
  - Interval estimates
- Remaining topics (special cases):
  - Estimate the mean
  - Prediction intervals
  - Difference between two means
  - Paired observations
  - Estimating the variance
  - Ratio of two variances
- Tools: CLT, Thm. 7.11, sampling theory.

Single Sample: Estimating the Mean

- Scenario: estimate μ.
- Point estimate of population mean: $\bar{X}$.
- Interval estimate of population mean: via Thm. 7.11 or CLT.

Estimate the Mean: How to Derive a CI

- Step 1: Set up probability statement capturing central $(1-\alpha)$ of Z.
- Step 2: Use
$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}.$$
- Step 3: Rearrange to isolate μ to obtain the confidence interval.

CI for μ (Case 1: σ^2 known)

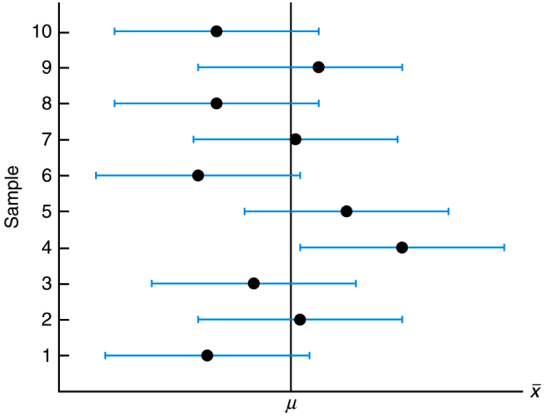
If $\bar{X}$ is the mean of a random sample of size n from a population with known variance σ^2, a $100(1-\alpha)\%$ CI for μ is:

$$\bar{x} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}} < \mu < \bar{x} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}},$$

where $z_{\alpha/2}$ leaves area $\alpha/2$ to the right.

Why do different samples give different CIs?

- Because $\bar{X}$ is a random variable: it varies from sample to sample.
- The CI endpoints are functions of $\bar{X}$ (and possibly S), so they also vary.



How do we interpret a confidence interval?

Suppose a 95% CI for μ is:

$$(12.3,\ 18.7)$$

Incorrect interpretation (very common):

- There is a 95% probability that μ is between 12.3 and 18.7.

Correct frequentist interpretation:

- If we repeatedly take samples and compute intervals the same way, about 95% of those intervals will contain the true mean μ.

Informal (acceptable) wording:

- Values of μ in this interval are plausible given the data and model.

Probability vs Confidence

Probabilities are used when reasoning from the _____ to the _____.

Confidence levels are used when reasoning from the _____ to the _____.

Options: population sample

Example: Keyboard Height (90% CI, σ known)

Engineers are trying to determine the preferred height for an ergonomic keyboard. A sample of $n = 31$ typists were selected and their preferred keyboard heights were determined. The resulting sample average preferred height was $\bar{x} = 80.0$ cm. Assuming that the preferred height is normally distributed with $\sigma = 2.0$ cm, obtain a CI for μ (the true average preferred height).

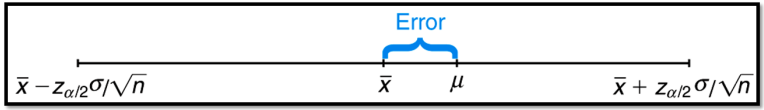
Engineers study preferred ergonomic keyboard height.

Error in Estimating μ

- Usually $\bar{x} \neq \mu$ so the point estimate has error.
- How big can the error be?

Theorem 9.1: If $\bar{x}$ is used to estimate μ, we can be $100(1-\alpha)\%$ confident the error does not exceed

$$z_{\alpha/2}\frac{\sigma}{\sqrt{n}}.$$



Determining Sample Size

We can pick n so that the error in estimating μ is less than ε.

Theorem 9.2: If $\bar{x}$ estimates μ, we can be $100(1-\alpha)\%$ confident the error will not exceed ε when

$$n = \left(\frac{z_{\alpha/2}\sigma}{\varepsilon}\right)^2.$$

Sample Size Example (Keyboard)

How large a sample is required to be 95% confident that the estimate of μ is off by less than 0.5 cm?

One-Sided Confidence Bounds (σ^2 known)

Sometimes we only want an upper or lower bound.
If $\bar{x}$ is the mean of a random sample size n with known σ^2:

- Upper one-sided $100(1-\alpha)\%$ bound:

$$\mu < \bar{x} + z_\alpha \frac{\sigma}{\sqrt{n}}$$

- Lower one-sided $100(1-\alpha)\%$ bound:

$$\mu > \bar{x} - z_\alpha \frac{\sigma}{\sqrt{n}}$$

CI for μ (Case 2: σ^2 unknown, normal)

If $\bar{x}$ and s^2 are from a normal population with unknown σ^2, a $100(1-\alpha)\%$ CI for μ is:

$$\bar{x} - t_{\alpha/2,\ \nu}\frac{s}{\sqrt{n}} < \mu < \bar{x} + t_{\alpha/2,\ \nu}\frac{s}{\sqrt{n}},$$

where $\nu = n - 1$ and $t_{\alpha/2,\nu}$ leaves area $\alpha/2$ to the right.

Example (Case 2): Synthetic Diamond Weights

Six diamond weights (karats):
0.46, 0.61, 0.52, 0.48, 0.57, 0.54.
Assume normal; find a 95% CI for μ.

CI for μ (Case 3: σ^2 unknown, large n)

If n is sufficiently large (typically $n \geq 30$), CLT implies $\bar{X}$ is approximately normal.
A $100(1-\alpha)\%$ CI for μ:

$$\bar{x} - z_{\alpha/2}\frac{s}{\sqrt{n}} < \mu < \bar{x} + z_{\alpha/2}\frac{s}{\sqrt{n}}.$$

Example (Case 3): Chemical Plant Yield

Daily yield recorded for $n = 50$ days: $\bar{x} = 871$ tons, $s = 21$ tons. Construct a 90% CI for μ.

Standard Error of a Point Estimate

- The *standard error* of an estimator is its standard deviation.
- For known σ CI:

$$\bar{x} \pm z_{\alpha/2}\frac{\sigma}{\sqrt{n}} = \bar{x} \pm z_{\alpha/2} \cdot \text{s.e.}(\bar{x})$$

- For unknown σ (normal):

$$\bar{x} \pm t_{\alpha/2}\frac{s}{\sqrt{n}} = \bar{x} \pm t_{\alpha/2} \cdot \text{s.e.}(\bar{x})$$

Prediction Intervals

- CI for μ quantifies uncertainty about the *mean*.
- Sometimes we want uncertainty for a *single future observation* X_0.
- This is a *prediction interval*.

Prediction Interval (Case 1: σ^2 known)

For normal measurements with unknown μ and known σ^2, a $100(1-\alpha)\%$ PI for a future observation x_0 is:

$$\bar{x} - z_{\alpha/2}\sigma\sqrt{1+\frac{1}{n}} < x_0 < \bar{x} + z_{\alpha/2}\sigma\sqrt{1+\frac{1}{n}}.$$

Prediction Interval (Case 2: σ^2 unknown)

For normal measurements with unknown μ and unknown σ^2, a $100(1-\alpha)\%$ PI for x_0 is:

$$\bar{x} - t_{\alpha/2,\ n-1}\, s\sqrt{1+\frac{1}{n}} < x_0 < \bar{x} + t_{\alpha/2,\ n-1}\, s\sqrt{1+\frac{1}{n}}.$$

Confidence Interval vs Prediction Interval: Hot Dogs

Fat content (%) of 10 hot dogs:
25.2, 21.3, 22.8, 17.0, 29.8, 21.0, 25.5, 16.0, 20.9, 19.5.
Assume normal.

- Find a 95% confidence interval for the mean fat content μ.
- Find a 95% prediction interval for the next hot dog.

Two Samples: Difference Between Two Means

- Scenario: estimate $\mu_1 - \mu_2$.
- Point estimate: $\bar{X}_1 - \bar{X}_2$.
- Sampling distribution (approx.): $\bar{X}_1 - \bar{X}_2$ is normal with

$$\mathbb{E}(\bar{X}_1 - \bar{X}_2) = \mu_1 - \mu_2, \qquad \mathrm{Var}(\bar{X}_1 - \bar{X}_2) = \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}.$$

CI for $\mu_1 - \mu_2$ (Case 1: σ_1^2, σ_2^2 known)

If $\bar{x}_1, \bar{x}_2$ are means of independent samples of size n_1, n_2 from populations with known variances σ_1^2, σ_2^2, then a $100(1-\alpha)\%$ CI for $\mu_1 - \mu_2$ is:

$$(\bar{x}_1 - \bar{x}_2) \pm z_{\alpha/2}\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}.$$

Example (Case 1): Car Ownership and GPA

An investigation is conducted to determine whether car ownership affects academic achievement. A random sample was drawn from population 1 (students that do NOT own cars) and an independent random sample was drawn from population 2 (students that own cars). Calculate a 95% Confidence Interval on $\mu_1 - \mu_2$, based on the data below. Given:

$$\sigma_1^2 = 0.36,\ n_1 = 100,\ \bar{x}_1 = 2.70; \qquad \sigma_2^2 = 0.40,\ n_2 = 100,\ \bar{x}_2 = 2.54.$$

Compute a 95% CI for $\mu_1 - \mu_2$.

CI for $\mu_1 - \mu_2$ (Case 2: equal but unknown variances)

If samples are independent, populations are approximately normal, and $\sigma_1^2 = \sigma_2^2$ but unknown, then a $100(1-\alpha)\%$ CI is:

$$(\bar{x}_1 - \bar{x}_2) \pm t_{\alpha/2,\ \nu}\, s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}},$$

where $\nu = n_1 + n_2 - 2$ and

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}.$$

Pooled Variance (Details)

When standardizing $\bar{X}_1 - \bar{X}_2$ under a common variance σ^2:

$$Z = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\sigma^2 \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}}.$$

We estimate σ^2 using the pooled estimator:

$$S_p^2 = \frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2}.$$

Example (Case 2): Ultrasonic Humidifiers

Brand 1 outputs (oz): 14.0, 14.3, 12.2, 15.1
Brand 2 outputs (oz): 12.1, 13.6, 11.9, 11.2

- What assumption(s) are needed to use pooled-t CI?
- Compute a 95% CI for $\mu_1 - \mu_2$.

CI for $\mu_1 - \mu_2$ (Case 3: unequal unknown variances)

If populations are approximately normal with $\sigma_1^2 \neq \sigma_2^2$ and both unknown, an approximate $100(1-\alpha)\%$ CI for $\mu_1 - \mu_2$ is:

$$(\bar{x}_1 - \bar{x}_2) \pm t_{\alpha/2,\ \nu} \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}},$$

with Welch–Satterthwaite degrees of freedom

$$\nu = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{\left(\frac{s_1^2}{n_1}\right)^2}{n_1 - 1} + \frac{\left(\frac{s_2^2}{n_2}\right)^2}{n_2 - 1}}.$$

Example (Case 3): Fabric Air Permeability

A study gives summary information on air permeability for 2 different types of plain-weave fabric (provided below). Assuming that the porosity distribution for both types of fabric is normal, calculate a 95% confidence interval for the difference between true average porosity for the fabrics.

Fabric	n	mean	sd
Cotton	10	51.71	0.79
Tracetate	10	136.14	3.59

Paired Observations

- Previously: two independent samples from independent populations.
- Now: one sample, but two measurements on each subject/item.
- These are *paired* data $\Rightarrow$ within-pair observations are not independent.

Paired Observations: Setup and Assumptions

- Data consist of n independent pairs (X_{1i}, X_{2i}).
- X_{1i} and X_{2i} are (approximately) normal.
- Define differences $d_i = X_{1i} - X_{2i}$.
- Assume d_i are normal with mean μ_D and variance σ_D^2.
- Point estimator: $\bar{D}$.
- To test/estimate mean difference, analyze the d_i's (paired t), not a two-sample test.

Paired Observations Concept Question

Which of the following situations involve paired observations? (Select all that apply.)
(a) Heights and weights measured for 50 students.
(b) Blood pressure of patients measured before and after taking a new medication.
(c) Exam scores from two different sections of a course taught by different instructors.
(d) Fuel efficiency of the same cars measured using gasoline type A and gasoline type B.
(e) Average household income from a random sample of Pittsburgh residents compared with a random sample of Philadelphia residents.

Paired t CI for $\mu_D = \mu_1 - \mu_2$

If $\bar{d}$ and s_d^2 are the mean and variance of the differences from n pairs, a $100(1-\alpha)\%$ CI for μ_D is:

$$\bar{d} - t_{\alpha/2,\ n-1}\frac{s_d}{\sqrt{n}} < \mu_D < \bar{d} + t_{\alpha/2,\ n-1}\frac{s_d}{\sqrt{n}}.$$

Paired Observations: Example (Slide vs Digital Retrieval)

13 medical professionals were timed retrieving an image via slides and via a digital database (Web front end).

Subject	1	2	3	4	5	6	7	8	9	10	11	12	13
Slide	30	35	40	25	20	30	35	62	40	51	25	42	33
Digital	25	16	15	15	10	20	7	16	15	13	11	19	19
Diff	5	19	25	10	10	10	28	46	25	38	14	23	14

- Let μ_D denote the true mean difference between slide retrieval time (sec) and digital retrieval time.
- Using the paired t confidence interval to estimate μ_D requires that the difference distribution be at least approximately normal.
- Calculate a 95% confidence interval for the true mean difference.

Paired Observations: Example (Compute 95% CI)

Single Sample: Estimating the Variance

- Scenario: estimate σ^2.
- Point estimate: S^2.
- For normal populations:

$$\frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{n-1}.$$

- Use this to build a confidence interval for σ^2.

CI for σ^2

If S^2 is the variance of a random sample of size n from a normal population, a $100(1-\alpha)\%$ CI for σ^2 is

$$\frac{(n-1)S^2}{\chi^2_{\alpha/2,\ n-1}} < \sigma^2 < \frac{(n-1)S^2}{\chi^2_{1-\alpha/2,\ n-1}},$$

where $\chi^2_{\alpha/2,\ n-1}$ and $\chi^2_{1-\alpha/2,\ n-1}$ leave right-tail areas $\alpha/2$ and $1-\alpha/2$, respectively.

Example: Concrete Strength Variance

A cement manufacturer claims compressive strength is stable (narrow range). A sample of $n = 10$ has $\bar{x} = 312$ and sample variance $s^2 = 195$ $(\text{kg}/\text{cm}^2)^2$. Find a 90% CI for σ^2.

Two Samples: Ratio of Two Variances

- To compare variability, estimate σ_1^2/σ_2^2.
- Point estimate: S_1^2/S_2^2.
- For normal populations with independent samples:

$$\frac{S_1^2/\sigma_1^2}{S_2^2/\sigma_2^2} \sim F_{n_1-1,\ n_2-1}.$$

- Use this to build a CI for σ_1^2/σ_2^2.

CI for σ_1^2/σ_2^2

If S_1^2 and S_2^2 are variances of independent samples of size n_1 and n_2 from normal populations, a $100(1-\alpha)\%$ CI for σ_1^2/σ_2^2 is

$$\frac{S_1^2/S_2^2}{f_{\alpha/2}(\nu_1,\nu_2)} < \frac{\sigma_1^2}{\sigma_2^2} < \frac{S_1^2/S_2^2}{f_{\alpha/2}(\nu_2,\nu_1)},$$

where $\nu_1 = n_1 - 1$, $\nu_2 = n_2 - 1$ and $f_{\alpha/2}(\nu_1,\nu_2)$ leaves right-tail area $\alpha/2$.

Example: Ratio of Variances (Crack Repair Polymers)

Data on impregnated compressive modulus (psi $\times 10^6$):

- Epoxy: 1.75, 2.12, 2.05, 1.97
- MMA prepolymer: 1.77, 1.59, 1.70, 1.69

Obtain a 90% CI for $\sigma^2_{\text{Epoxy}}/\sigma^2_{\text{MMA}}$.

Which interval do I use?

1. What are you estimating?
   - Mean $\mu \Rightarrow$ CI for μ
   - Future observation $X_0 \Rightarrow$ Prediction Interval
   - Difference $\mu_1 - \mu_2 \Rightarrow$ Two-sample CI
   - Paired mean $\mu_D \Rightarrow$ Paired t
   - Variance $\sigma^2 \Rightarrow \chi^2$ CI
   - Ratio $\sigma_1^2/\sigma_2^2 \Rightarrow F$ CI
2. Data structure
   - One sample
   - Two independent samples
   - Paired data $\Rightarrow$ analyze differences d_i
3. Which distribution?
   - σ known $\Rightarrow z$ (rare)
   - σ unknown $\Rightarrow t$

Two-sample variance choice

- Equal variances $\Rightarrow$ pooled t
- Unequal $\Rightarrow$ Welch t (default)

Big picture

- CI = parameter uncertainty
- PI = future observation uncertainty